
Learning with complex problems

Ensure a solid foundation of your project by keeping complex problems at the centre during all steps for the design and implementation process of the transdisciplinary education format. Below, you will learn how to choose and use the right challenges to help both students and partners develop competences needed to contribute to societal solutions.

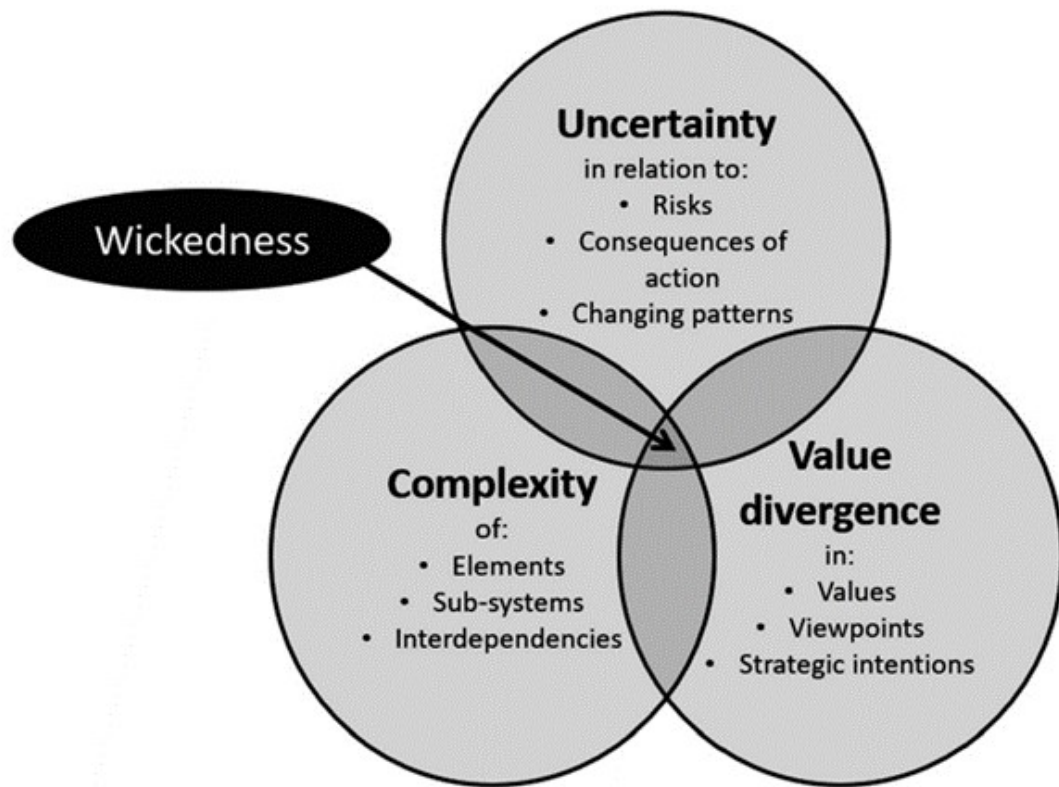
The complex problem at the heart of transdisciplinary education forms the context in which you learn. When you focus on a complex problem, you work on your professional development and on societal or sustainability goals. Such as learning to deal with uncertainty and complexity (Horn et al., 2023 and Veltman et al., 2021).

Complex problems, or ‘wicked problems’, have 3 characteristics:

1. Complexity: wicked problems consist of multiple elements that are interconnected with each other. This makes these problems complex.

2. Uncertainty: this aspect deals with the uncertainties about risks, consequences of actions and changes that occur over time.

3. Value differences: different interests, perspectives and values are at play among the parties involved. This can lead to differing views on what is important and how to address the problem.



(Veltman et al., 2021)

Make sure the complex problem meets these characteristics. This allows students and partners to develop the competences they need to contribute to complex social problems.

In addition, there are other criteria or considerations for selecting a wicked problem:

1. When selecting a wicked problem, it is important that it involves real partners who also have an interest in the problem (Groothuijsen, 2021).
2. Both the academic partners (lecturers, students, researchers) and other partners (field, government, etc.) can come up with a wicked problem. Dwell on how this affects collaboration and impact (Groothuijsen, 2021).

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3. The transdisciplinary group itself can choose a wicked problem based on a general theme. This has advantages over a predetermined problem, such as greater intrinsic motivation. But there are also disadvantages to this choice, such as the impact the project has and the length of time to arrive at the wicked problem. These advantages and disadvantages should be carefully weighed against each other.

Tool: The [complex-tool](#)

[in Dutch] helps with formulating the issue in such a way that the formulation of a social problem fits the learning 'level' of those involved.